

Diagnostic information - DME (DFI) control unit

Main relay off

Diagnostic conditions

- Vehicle voltage > 8.0 V
- Ignition off
- Control unit run-on completed (see ⓘ)
- Then ignition on

Possible fault causes

- ♦ Main relay faulty (contact stuck)
- ♦ Short circuit to B+ in line between main relay and DME (DFI) control unit (DME (DFI) control unit power supply)
- ♦ Short circuit to ground in line between DME (DFI) control unit and main relay (main relay activation)

Fault setting condition(s)

- Power supply for DME (DFI) control unit > 4.0 V at connector [B], contact 3/5/6, with:
- Power supply for DME (DFI) control unit is 0.0 V at connector [B], contact 50 (terminal 15)

ⓘ INFORMATION

After the ignition is switched off (terminal 15), the DME control unit switches to control unit run-on mode.

- ♦ The run-on period depends on many underlying conditions (e.g. engine temperature for fan activation).
- ♦ The run-on duration can be 1 minute when the engine is cold and up to 25 minutes when the engine is hot.

P0687

Diagnostic information - DME (DFI) control unit

Main relay ON

Diagnostic conditions

- Vehicle voltage > 8.0 V
- Ignition on

Possible fault causes

- ♦ Main relay faulty
- ♦ Open circuit in power supply for main relay control circuit
- ♦ Short circuit to ground or open circuit in the lines between the main relay and DME (DFI) control unit (DME (DFI) control unit power supply)
- ♦ Short circuit to B+ or open circuit in the lines between the DME (DFI) control unit and main relay (main relay activation)

Fault setting condition(s)

- Power supply for DME (DFI) control unit < 6.0 V at connector [B], contact 3/5/6, with:
 - Power supply for DME (DFI) control unit > 8.0 V at connector [B], contact 50 (terminal 15)

P0698

Diagnostic information - DME (DFI) control unit

Sensor voltage 3

Diagnostic conditions

- Ignition on > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Component faulty (short circuit)

Connector [A], contact 48:

- ♦ Crankshaft sensor

Fault setting condition(s)

- Deviation in sensor voltage:
 - ♦ < 4.7 V (P0698)
 - or
 - ♦ > 5.3 V (P0699)

P0699

Diagnostic information - DME (DFI) control unit

Sensor voltage 3

Diagnostic conditions

- Ignition on > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Component faulty (short circuit)

Connector [A], contact 48:

- ♦ Crankshaft sensor

Fault setting condition(s)

- Deviation in sensor voltage:
 - ♦ < 4.7 V (P0698)
 - or
 - ♦ > 5.3 V (P0699)

P0700

Diagnostic information - DME (DFI) control unit

Engine warning light **[Check Engine]** on due to fault in transmission control unit

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Fault entry in PDK control unit (see )

INFORMATION

Different fault healing strategies in the engine and transmission control units can lead to a situation in which the original transmission fault no longer exists in the transmission fault memory (healed) but the fault (P0700) nevertheless remains entered in the engine control unit (not yet healed).

P0701

Diagnostic information - DME (DFI) control unit

Torque intervention from transmission control unit implausible

Diagnostic conditions

- Drive with gear changes

Possible fault causes

- ♦ Shifting in PDK takes too long

Fault setting condition(s)

- The torque intervention by the PDK control unit takes too long; it is therefore implausible for the DME control unit.

P0705

Diagnostic information - DME (DFI) control unit

Selector-lever position implausible

Diagnostic conditions

- Drive with gear changes

Possible fault causes

- ♦ Selector-lever position is reported as implausible

P0716

Diagnostic information - DME (DFI) control unit

Manual transmission speed sensor: difference from engine speed

Diagnostic conditions

- Time since engine start > 5 second(s)
- Clutch closed

Possible fault causes

- ♦ Clutch plate slipping
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Manual transmission speed sensor loose
- ♦ Rpm sensor wheel in transmission faulty

Fault effects

- ♦ Reduced engine torque

P0717

Diagnostic information - DME (DFI) control unit

Manual transmission speed sensor: implausible or no signal

Diagnostic conditions

- Time since engine start > 5 second(s)
- Clutch closed

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Manual transmission speed sensor loose
- ♦ Manual transmission speed sensor faulty
- ♦ Rpm sensor wheel in transmission faulty

Fault effects

- ♦ Reduced engine torque

P0718

Diagnostic information - DME (DFI) control unit

Manual transmission speed sensor: implausible or no signal

Diagnostic conditions

- Time since engine start > 5 second(s)
- Clutch closed

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Manual transmission speed sensor loose
- ♦ Manual transmission speed sensor faulty
- ♦ Rpm sensor wheel in transmission faulty

Fault effects

- ♦ Reduced engine torque

P0832

Diagnostic information - DME (DFI) control unit

Cruise control clutch switch (upper)

Diagnostic conditions

- Drive with at least 40 gear changes

Possible fault causes

- ♦ Gear change without clutch actuation
- ♦ Short circuit to B+ in cruise control clutch switch (upper) - DME (DFI) control unit
- ♦ Cruise control clutch switch (upper) faulty

Fault setting condition(s)

- 20 gear changes, with:
 - Clutch switch interlock (lower) indicates clutch NOT actuated

or

- 40 gear changes, with:
 - Clutch switch interlock (lower) indicates clutch actuated

Fault effects

- ♦ Start Stop system does not work

P0834

Diagnostic information - DME (DFI) control unit

Clutch switch interlock (lower)

Diagnostic conditions

- Ignition on and clutch pedal fully depressed
- None of the following faults stored: P0830

Possible fault causes

- ♦ Short circuit to ground in signal line
- ♦ Clutch switch interlock (lower) faulty (stuck closed)

Fault setting condition(s)

- Clutch switch interlock (lower) actuated with:
 - Clutch switch for cruise control (upper) not actuated

Fault effects

- ♦ Engine start possible, with:
 - Clutch pedal not pressed

P0835

Diagnostic information - DME (DFI) control unit

Clutch switch interlock (lower)

Diagnostic conditions

- When engine start is detected
and
- Clutch pedal fully depressed

or

- Starter does not turn
and
- Following sequence 3 x in series:
 - Clutch pedal fully depressed
 - Ignition lock position 2 (terminal 50 - engine start) > 3 second(s) (> 9 seconds in total)
 - then:
 - Ignition off
and
 - Release clutch pedal

Possible fault causes

- ♦ Short circuit to B+ or open circuit in signal line
- ♦ Clutch switch interlock (lower) faulty (stuck open)

Fault setting condition(s)

- Ignition lock position 2 (terminal 50 - engine start) > 9 second(s) in total (See Diagnostic conditions)

Fault effects

- ♦ Starter does not turn
- ♦ Display in instrument cluster > [Press clutch pedal](#) <

P083F

Diagnostic information - DME (DFI) control unit

Clutch switch implausible in relation to clutch travel sensor (parking brake control unit - EPB)

Diagnostic conditions

- Ignition on
- Completely depress the clutch pedal once and release it again
- None of the following faults stored: P0830, P0831, P0834, P0835, U0128, U0417

Possible fault causes

- ♦ Fault in clutch travel sensor (connected to parking brake control unit) WITHOUT fault entry
- ♦ Short circuit to B+ in clutch switch interlock (lower) - DME (DFI) control unit
- ♦ Cruise control clutch switch (upper) faulty
- ♦ Short circuit to B+ in clutch switch interlock (lower) - DME (DFI) control unit
- ♦ Clutch switch interlock (lower) faulty

Fault setting condition(s)

 The signals for the upper clutch switch (cruise control clutch switch (upper)) and lower clutch switch (clutch switch interlock (lower)) are compared with the clutch pedal travel (parking brake control unit ► CAN ► SG_dfi).

- The fault is entered when the clutch pedal is fully depressed,
if the upper switch switches too late (compared to the travel signal),
or
if the lower one switches too early (compared to the travel signal).
- The fault is entered when the clutch pedal is released,
if the upper switch switches too early (compared to the travel signal),
or
if the lower one switches too late (compared to the travel signal).

Fault effects

- ♦ Start/Stop system does not function

P0850

Diagnostic information - DME (DFI) control unit

P/N input signal, open circuit/short circuit to B+

Diagnostic conditions

- Ignition on > 3 second(s)
- then start engine, with:
 - Brake pressed
 - Selector-lever position P or N

Possible fault causes

- ♦ Open circuit/short circuit to B+ in signal line

Fault setting condition(s)

Deviation in signal for P/N input signal and signal via CAN

Fault effects

- ♦ Starter does not turn

P0851

Diagnostic information - DME (DFI) control unit

P/N input signal, short circuit to ground

Diagnostic conditions

- Ignition on > 3 second(s)
- then start engine, with:
 - Brake pressed
 - Selector-lever position P or N

Possible fault causes

- ♦ Short circuit to ground in signal line

Fault setting condition(s)

Deviation in signal for P/N input signal and signal via CAN

Fault effects

- ♦ Engine start possible, with:
 - Selector-lever position R or D or M

Diagnostic information - DME (DFI) control unit

Torque request from PSM implausible

Diagnostic conditions

- Drive with PSM intervention for vehicle stabilization

Possible fault causes

- ♦ Fault in PSM control unit area

Fault setting condition(s)

- Torque request from PSM is implausibly high compared to driver setpoint torque (via accelerator pedal).

Fault effects

- ♦ All torque requests from the PSM are ignored for this trip ⇒ PSM interventions restricted.

Diagnostic information - DME (DFI) control unit

Fuel pressure/quantity too low at start

INFORMATION 1

This fault entry can be caused by various reasons.

- ♦ If the fault was **NOT entered while the engine was starting**, but while the engine was running normally, a check must be carried out in the **crankshaft sensor** area (**See  2**).
- ♦ Fault finding in the fuel area must only be carried out if a customer complains that starting times are too long AND if the fault was entered while the engine was starting.
- ♦ If there is no such customer complaint, however, the only measure to be taken is to clear the fault memory.

Diagnostic conditions

- Engine start
- Engine temperature - 10 ... +100 °C (14 ... 212 °F)
- Engine running > 5 second(s)

Possible fault causes

- ♦ Short-term failure of the engine speed signal as a result of a crankshaft sensor fault with or without a corresponding fault entry
(See  2)
- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
(See  3)
- ♦ Quantity control valve faulty (in fuel high-pressure pump)
- ♦ Fuel high-pressure pump faulty

Fault setting condition(s)

- The fuel pressure required for carrying out a high-pressure start was not reached or was not reached quickly enough
 -  Low-pressure start is enabled; starting takes considerably longer (up to 2 sec.).

INFORMATION 2

- If engine speed = 0, a starting setpoint value is set for the fuel high pressure that is higher than the setpoint value at idle speed or at a low load.
- An engine start is detected if the engine speed increases from 0 to > 300 rpm.
- When an engine start is detected, a check determines whether the measured fuel high pressure reaches the setpoint value within a certain number of engine revolutions.
⇒ If the engine's speed signal fails briefly (e.g. 6/1,000 seconds) and is then restored, the following things happen:
 1. **Engine speed = 0** ► Starting setpoint value for the fuel high pressure is set.
 2. **Engine speed increases to over 300 rpm** ► Engine start is detected.
 3. **Comparison of setpoint value and actual value of the fuel high pressure** ► Fault entry P1023, although there was no actual engine start.
 4. **Entries in the extended fault memory for P1023 are as follows:**
 - ♦ 40_Time after Start > 5 s
 - ♦ 40_Engine RPM > 600 rpm

♦ 40_ *Vehicle Speed* > 0 km/h (0 mph)

i INFORMATION 3

Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).

♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

P1024

Diagnostic information - DME (DFI) control unit

Fuel high pressure too high after engine off

Diagnostic conditions

- Engine temperature - 10 ... +100 °C (14 ... 212 °F)
- Engine running > 5 second(s)
- then ignition off > 10 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines ► Fuel high-pressure sensor
- ♦ Quantity control valve faulty (in fuel high-pressure pump)

Fault setting condition(s)

- Fuel high pressure drops too slowly after engine off

Diagnostic information - DME (DFI) control unit

Fuel high pressure adaptation

INFORMATION 1

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

Diagnostic conditions

- Engine speed 450 ... 4,000 rpm
- Engine temperature - 10 ... + 100 °C (14 ... 212 °F)
- Driving with different operating conditions

Possible fault causes

- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
- ♦ Fuel high-pressure adaptation required (see  2)
- ♦ Fuel high-pressure pump faulty

INFORMATION 2

The old values must be deleted prior to a new adaptation of the fuel high-pressure ranges.

- ♦ The fuel high-pressure adaptation values can be deleted using the **PIWIS Tester** as follows:

DME ⇒ Maintenance/repairs ⇒ Delete adaptation values

- ♦ The new adaptation takes place while driving.

P1027

Diagnostic information - DME (DFI) control unit

Fuel high pressure fluctuates

Diagnostic conditions

- Engine speed 450 4,000 rpm
- Engine temperature - 10 ... +100 °C (14 ... 212 °F)
- Driving with different operating conditions

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines ► Fuel high-pressure sensor
- ♦ Quantity control valve sluggish (in fuel high-pressure pump)

Fault setting condition(s)

The fuel high pressure fluctuates by +/- 25 bar from the setpoint fuel pressure.

Diagnostic information - DME (DFI) control unit

Fuel high-pressure sensor stuck at engine start

INFORMATION

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

Diagnostic conditions

- Engine start
- Engine temperature - 10 ... +100 °C (14 ... 212 °F)

Possible fault causes

- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
- ♦ Fuel high-pressure sensor faulty

Fault setting condition(s)

No change in the **fuel high-pressure sensor** voltage line, at:

- Engine start

Diagnostic information - DME (DFI) control unit

Fuel high-pressure pump: bank 1 adaptation fault

Diagnostic conditions

- Perform system test >Fuel high-pressure pump, bank 1<
- This adaptation is also performed during a sufficiently long period of steady driving for the Macan only; for all other vehicles, this adaptation is performed only during the system test.

INFORMATION

If the system test takes longer than approx. 5 minutes, it will be cancelled without a result.

- This is the case if the tested fuel high-pressure pump can no longer build up high pressure, i.e. can no longer pump fuel (e.g. low-pressure supply blocked, pump tappets broken, pump faulty, ...).

Possible fault causes

- ♦ Bank 1 fuel high-pressure pump faulty
- ♦ Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see  below)

Fault setting condition(s)

 During the >Fuel high-pressure pump, bank 1< system test, the fuel high-pressure pump for bank 2 is adjusted to zero quantity (switched off) and the fuel high-pressure pump for bank 1 is adapted.

- A fault is entered if the result is not within the permissible adaptation range.

Diagnostic information - DME (DFI) control unit

Fuel high-pressure pump: bank 2 adaptation faulty

Diagnostic conditions

- Perform system test >Fuel high-pressure pump, bank 2<

INFORMATION

If the system test takes longer than approx. 5 minutes, it will be canceled without a result.

- This is the case if the tested fuel high-pressure pump can no longer build up high pressure, i.e. can no longer pump fuel (e.g. low-pressure supply blocked, pump tappets broken, pump faulty, ...).

Possible fault causes

- ♦ Fuel high-pressure pump, bank 2 faulty

Fault setting condition(s)

 During the >Fuel high-pressure pump, bank 2< system test, the fuel high-pressure pump for bank 1 is adjusted to zero quantity (switched off) and the fuel high-pressure pump for bank 2 is adapted.

- A fault is entered if the result is not within the permissible adaptation range.

P1044

Diagnostic information - DME (DFI) control unit

Acoustic simulator 1 solenoid valve output stage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Acoustic simulator 1 solenoid valve faulty

P1045

Diagnostic information - DME (DFI) control unit

Acoustic simulator 1 solenoid valve output stage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Acoustic simulator 1 solenoid valve faulty

P1046

Diagnostic information - DME (DFI) control unit

Acoustic simulator 1 solenoid valve output stage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Acoustic simulator 1 solenoid valve faulty

P1047

Diagnostic information - DME (DFI) control unit

Acoustic simulator 2 solenoid valve output stage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Acoustic simulator 2 solenoid valve faulty

P1048

Diagnostic information - DME (DFI) control unit

Acoustic simulator 2 solenoid valve output stage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Acoustic simulator 2 solenoid valve faulty

P1049

Diagnostic information - DME (DFI) control unit

Acoustic simulator 2 solenoid valve output stage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line (power supply or activation)
- ♦ Acoustic simulator 2 solenoid valve faulty

P1058

Diagnostic information - DME (DFI) control unit

Tank vent pressure sensor implausible

Diagnostic conditions

- Engine off
- Ignition on
- Only slight change in ambient pressure (A060) since last ignition on
- Fault P1559 not entered

Possible fault causes

- ♦ Fault entry P1559 - Control unit restart during operation (warm reset)

① If the fault P1559 is entered, the fault P1058 can also be entered although everything is OK in the tank ventilation area.

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Ground supply faulty
- ♦ Tank vent pressure sensor faulty
- ♦ Tank vent valve was stuck closed the last time the engine was stopped

Fault setting condition(s)

- Difference between tank vent pressure sensor and ambient pressure sensor is more than 135 hPa.

P1084

Diagnostic information - DME (DFI) control unit

Signal from boost pressure adjuster, bank 1, faulty

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines (signal line feedback)
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1085

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 1: electrical fault

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1086

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 1: electrical fault

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1087

Diagnostic information - DME (DFI) control unit

Signal from boost pressure adjuster, bank 2, faulty

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines (signal line feedback)
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1088

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 2: electrical fault

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1089

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 2: electrical fault

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1147

Diagnostic information - DME (DFI) control unit

Oil pressure control valve: power supply

INFORMATION

- ♦ The oil pressure control valve is supplied with 12 V while the engine is running.
- ♦ A short circuit to B+ can be detected only during control unit run-on.

Diagnostic conditions

- Engine running, then:
- Ignition off > 1 minute(s)

Possible fault causes

- ♦ Short circuit to B+ (power supply to oil pressure control valve)
- ♦ Oil pressure control valve faulty

P1148

Diagnostic information - DME (DFI) control unit

Oil pressure control valve: power supply

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Open circuit/short circuit to ground (power supply to oil pressure control valve)
- ♦ Short circuit to B+/ground or open circuit in the line (oil pressure control valve activation)
- ♦ Oil pressure control valve faulty